Bring us the hard problem

Wyspa Technologies is a US small business in Newport, RI, doing naval architecture and engineering for fast small craft since 2018: 3D laser scanning, structural analysis, composite structures, hydrofoils and appendages, and speed-power prediction. We now apply this work to unmanned surface vessels (USVs).

ENTITY	UEI	CAGE	BUSINESS TYPE
Wyspa Technologies LLC	NWJRCE69AWH9	Pending	Small business · since 2018

Core competencies

- **3D laser scanning and as-built models.** Our own FARO scanner, registered point clouds and CAD models. Scanning fed engineering on 101 projects.
- **Structural analysis and composite structures.** FEA on 31 projects (Femap/Nastran, SolidWorks Simulation), laminate schedules (ply books), and ply flattening to DXF cut files.
- **Hydrofoils and appendages.** Foils, rudders and keels on 64 projects, sized to cavitation and ventilation limits.
- **Speed-power and range at speed.** In-house prediction tools for hulls and hydrofoils, checked against ORC VPP and WinDesign on sailing craft.
- **Stability and USCG compliance.** Hydrostatics, stability and incline (GHS, Orca3D), and reports for USCG review under 46 CFR Subchapter T.
- **Ride control (simulation).** Foil-borne motion simulation with active control of ride height, pitch and roll. Not yet tested on the water.

Concept: hydrofoil USV

A 20 ft class unmanned hydrofoil craft with a proa hull.

DESIGN TARGETS	
Payload	500 lb
Transit speed, foil-borne	20 kn
Range at 20 kn, 500 lb	30 nm electric; about 350 nm hybrid
Operation, foil-borne	Waves up to about 0.3 m (Sea State 2) as drawn; Sea State 3 needs deeper foils (under study)

Concept design targets, from calculation. Not yet built or tested.

How to work with us

We take subcontract work packages, surge engineering and small purchase orders from USV builders, primes and government offices.

Why Wyspa

- Scanning, modeling, analysis and build-ready files in one firm. No hand-off between a scan vendor and an engineering firm.
- Our own speed-power and laminate tools, so studies turn around quickly.
- A 19 ft hull available for a low-cost testbed, to check designs on the water.

Relevant experience (commercial)

- **USCG Subchapter T conversions.** A historic 12 Metre (USCG approved) and a 12-passenger Pearson 35 (structure approved, rig review pending). Scans on both; inclining test, stability and rig-loads reports on the Pearson.
- **85 ft aluminum patrol boat** (2026). Scanned on site at a working repair yard. As-built model of the outer surfaces in progress.
- **Hydrofoils for a 43 ft powercat**, with CFD of the foil system.
- **Electric-aircraft program:** 3D scans of fuselage and wings, and composite manufacturing support.
- **90+ racing-yacht projects** (TP52, Swan 42, Etchells): performance prediction, rating optimization, keels, rudders and laminates.

NAICS

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